

Qubits as Unit Vectors in $\mathbb{C}^2$ and the Born Rule as a Probability Measure

Schola Arcana · Proof Vault · Track 2 — Quantum Computing

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Motivation

Classical computation operates on bits — definite values in $\{0, 1\}$. Quantum computation generalises this by allowing *superpositions*: linear combinations of basis states weighted by complex amplitudes. The mathematical object encoding such a superposition is a unit vector in $\mathbb{C}^2$, and the rule governing what we observe upon measurement is the Born rule.

This entry has two goals. First, to establish formally that a qubit is a unit vector in $\mathbb{C}^2$ and to identify the geometric structure this imposes. Second, to verify that the Born rule satisfies the Kolmogorov axioms for a probability measure on the outcome space $\{0, 1\}$, grounding quantum probability in classical probability theory.

1 Part I — Qubits as Unit Vectors in $\mathbb{C}^2$

The Classical Bit

A classical bit is an element of $\{0, 1\}$ — a definite, discrete value. At any moment it occupies exactly one state. There is no notion of superposition; the two states are mutually exclusive.

The Computational Basis

We work over the field $\mathbb{C}$ of complex numbers. Define the *computational basis* of $\mathbb{C}^2$ as the two standard basis vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These are orthonormal: $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = 0$.

Definition 1 (Qubit). A *qubit* is a unit vector in $\mathbb{C}^2$, written in Dirac notation as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

where $\alpha, \beta \in \mathbb{C}$ are called *amplitudes*, subject to the *normalisation condition*:

$$\langle \psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

Remark 1. The Dirac notation $|\cdot\rangle$ denotes a column vector (*ket*), and $\langle\cdot|$ denotes its conjugate transpose (*bra*), so that $\langle\psi|\psi\rangle = |\psi\rangle^\dagger |\psi\rangle$ is the standard Hermitian inner product on $\mathbb{C}^2$.

The Geometry of the Qubit State Space

Proposition 1 (Qubit states form the 3-sphere). *The set of all qubit states is the unit sphere $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$.*

Proof. Write $\alpha = a_1 + ia_2$ and $\beta = b_1 + ib_2$ with $a_1, a_2, b_1, b_2 \in \mathbb{R}$. Then:

$$|\alpha|^2 + |\beta|^2 = a_1^2 + a_2^2 + b_1^2 + b_2^2 = 1.$$

This is precisely the equation of the unit sphere S^3 in $\mathbb{R}^4 \cong \mathbb{C}^2$. Hence the set of normalised qubit states is S^3 . $\square$

Remark 2 (The Bloch sphere). In practice, the global phase of a qubit is physically unobservable: states $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ yield identical measurement statistics for all observables. Quotienting S^3 by this $U(1)$ phase freedom via the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$ yields the *Bloch sphere* S^2 , the standard geometric representation of a single qubit's physically distinct states. Every point on S^2 corresponds to a unique physical state, parametrised as:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \varphi \in [0, 2\pi).$$

Example 1 (The computational basis states on the Bloch sphere). Setting $\theta = 0$ gives $|0\rangle$ (the north pole of S^2). Setting $\theta = \pi$ gives $|1\rangle$ (the south pole). Setting $\theta = \pi/2, \varphi = 0$ gives the equal superposition $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ (on the equator).

2 Part II — The Born Rule as a Probability Measure

The Kolmogorov Axioms

Recall that a *probability measure* on a sample space Ω with event algebra $\mathcal{F}$ is a function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying:

(K1) **Non-negativity:** $P(A) \geq 0$ for all $A \in \mathcal{F}$.

(K2) **Normalisation:** $P(\Omega) = 1$.

(K3) **Countable additivity:** For pairwise disjoint events $A_1, A_2, \dots \in \mathcal{F}$: $P(\bigcup_i A_i) = \sum_i P(A_i)$.

The Born Rule

Definition 2 (Measurement in the computational basis). Given a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, a *measurement in the computational basis* yields outcome 0 or 1 according to the *Born rule*:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

After measurement, the qubit *collapses* to $|0\rangle$ or $|1\rangle$ corresponding to the observed outcome.

Theorem 2 (The Born rule is a probability measure). Let $\Omega = \{0, 1\}$ with the power set $\mathcal{F} = 2^\Omega$, and let $P : \mathcal{F} \rightarrow \mathbb{R}$ be defined by the Born rule. Then P is a probability measure on $(\Omega, \mathcal{F})$.

Proof. We verify the three Kolmogorov axioms.

(K1) Non-negativity. For any $z \in \mathbb{C}$, we have $|z|^2 = z \cdot \bar{z} \geq 0$. Hence $P(0) = |\alpha|^2 \geq 0$ and $P(1) = |\beta|^2 \geq 0$. For the empty event: $P(\emptyset) = 0$ by convention. For $\{0\}$ and $\{1\}$: non-negativity holds as shown. For $\Omega = \{0, 1\}$: $P(\Omega) \geq 0$ follows from $|\alpha|^2 + |\beta|^2 = 1 > 0$. $\checkmark$

(K2) Normalisation. The sample space $\Omega = \{0, 1\}$ is the certain event. By the normalisation condition on the qubit state:

$$P(\Omega) = P(\{0, 1\}) = P(0) + P(1) = |\alpha|^2 + |\beta|^2 = 1. \checkmark$$

(K3) Countable additivity. Since $\Omega = \{0, 1\}$ is finite, it suffices to verify finite additivity. The only non-trivial partition of Ω into disjoint events is $\{0\}$ and $\{1\}$. We have:

$$P(\{0\} \cup \{1\}) = P(\Omega) = 1 = |\alpha|^2 + |\beta|^2 = P(\{0\}) + P(\{1\}).$$

All other cases (involving $\emptyset$) are immediate. Countable additivity holds since the space is finite. ✓

Therefore P satisfies all three Kolmogorov axioms and is a probability measure on $(\{0, 1\}, 2^{\{0,1\}})$. ■ □

What the Born Rule Does Not Say

Remark 3 (Amplitudes are not probabilities). The amplitudes $\alpha, \beta \in \mathbb{C}$ are *not* probabilities themselves. They can be negative, imaginary, or complex, and can interfere destructively — amplitudes from different computational paths can cancel each other out. This is the key feature absent in classical probability theory. The probabilities $|\alpha|^2$ and $|\beta|^2$ are derived from the amplitudes via the modulus-squared map, which discards phase information. This is why quantum amplitudes are sometimes called *probability amplitudes*: they are the square roots of probabilities, extended to the complex plane.

Remark 4 (The measurement problem). The Born rule is a postulate — it is not derived from more fundamental principles within standard quantum mechanics. Why physical measurement yields probabilities according to $|\alpha|^2$ rather than some other function of the amplitude remains one of the deepest unresolved questions in the foundations of quantum mechanics. The measurement problem — why and how a superposition collapses to a definite outcome — is genuinely open and connects directly to the interpretation debates (Copenhagen, many-worlds, QBism, relational QM) discussed elsewhere in these notes.

The Full Picture

A qubit is a unit vector in $\mathbb{C}^2$, parametrising a three-sphere S^3 of states (or a two-sphere S^2 once global phase is quotiented out). The classical bits 0 and 1 are the special cases $|0\rangle$ and $|1\rangle$ — the poles of the Bloch sphere. Everything in between is superposition: a genuinely quantum phenomenon with no classical analogue.

The Born rule assigns probabilities to measurement outcomes via $|\alpha|^2$ and $|\beta|^2$. We have verified that these assignments form a proper probability measure in the sense of Kolmogorov, grounding quantum probability in the same axiomatic foundation as classical probability theory. The crucial difference is that the amplitudes α, β themselves are complex and can interfere — a structure richer than classical probability, which uses only non-negative reals.

This richer structure is precisely what quantum algorithms exploit. Interference allows wrong answers to cancel and right answers to reinforce — the fundamental mechanism underlying Grover’s algorithm, Shor’s algorithm, and quantum computation in general.

One Question That Remains Open For Me

The Born rule is a postulate. But in the many-worlds interpretation, where no collapse occurs, the Born rule must be derived from rationality constraints on agents in branching universes (Deutsch-Wallace decision-theoretic argument). Does this derivation actually work — does it genuinely justify why a rational agent should assign credences equal to $|\alpha|^2$? And what does this mean for the status of probability in a deterministic universe?

